

07/04/25

ERIDAN company

Sunnyvale, CA.

Dear Hiring Team at **ERIDAN**

I am writing to express my interest in the Embedded Software Engineer position at ERIDAN. With over five years of experience in embedded systems development, particularly in the automotive and railway industries, I have background in C/C++ programming, AUTOSAR architecture, real-time systems, and hardware-software integration.

In my most recent role at BorgWarner, I collaborated with global cross-functional teams to ensure successful software integration and deployment, using version control tools such as Git and Plastic.

At CIDEC, I expanded my expertise in AUTOSAR by developing application-layer software components in C and configuring low-level drivers within the Hardware Abstraction Layer (HAL). I performed extensive debugging using JTAG and Ozone to resolve software issues and participated in the full software development lifecycle—from requirements analysis to integration testing.

Earlier in my career at CIDESI, I developed low-level drivers, implemented control algorithms using RTOS (SafeRTOS and FreeRTOS), performed reverse engineering, and conducted field testing on railway control systems. I also gained experience with embedded Linux and Python through projects involving Raspberry Pi platforms.

I hold a Bachelor's degree in Mechatronics Engineering and a Master's in Mechatronics, during which I worked on embedded data acquisition systems using FPGAs and microcontrollers. My technical knowledge, combined with hands-on experience and strong analytical skills, makes me a strong candidate for contributing to innovative embedded software projects at ERIDAN.

I would welcome the opportunity to further discuss how my background aligns with your team's needs. Thank you for considering my application.

Sincerely,

Ivonne Berenice Lemus Martínez.

+14089903068 | iberenicelemus@gmail.com